

EDUCATION

University of Massachusetts Amherst <i>Master of Science in Biostatistics, GPA 4.0</i> Relevant Courses: Multivariate Calculus, Linear Algebra I&II, Bayesian Modeling, Causal Inference, Statistical Methods for Data Science	Amherst, MA May 2025
University of Massachusetts Amherst <i>Bachelor of Science in Public Health, GPA 3.7</i>	Amherst, MA May 2024

EXPERIENCE

Iron Eagle X Defense Technology & AI solutions <i>Data Science Intern</i>	Tampa, FL Jan 2024 – Dec 2024
- Developed and deployed AI/ML models (e.g., deep learning, NLP) using PyTorch, TensorFlow, and scikit-learn to analyze large-scale, real-world datasets. - Engineered and productionized end-to-end ML pipelines, integrating Flask APIs, Docker, optimized for cloud environments. - Collaborated on high performance computing image analytics projects.	
Prapela Pediatric Medical Device Company <i>Statistical Consultant</i>	Biddeford, ME Sep 2023 – Feb 2025
- Advised on FDA De Novo submission that secured clearance for a pediatric device with Breakthrough Designation. - Performed statistical analysis of sensor data to validate stimulation equivalency across clinical and production devices. - Applied statistical modeling (Python, R) to support regulatory strategy and experimental design.	
Dimensional Insight Business Intelligence and Analytics Software <i>Business Intelligence Analyst Intern</i>	Burlington, MA May 2023 – August 2023
- Designed and implemented analytical dashboards to translate complex data insights into information suitable for non-technical hospital administrators. - Integrated advanced business logic into data ingestion and processing pipelines, reconciling large-scale datasets.	

PROJECTS

Covid Outcome Bayesian Modeling
Developed Bayesian hierarchical GLMs to quantify hospitalization risk from COVID-19, using spatial statistics, hypothesis testing, and causal inference to uncover risk disparities by geography and race.
Length of Stay and Disease Severity Prediction
Built a combined cohort regression model to predict disease severity and length of stay using synthetic claims data that achieved MAE of 0.149 days outperforming existing literature based models by at least 2 days.

SKILLS

Programming: Python, R, SQL, Bash, Pandas, NumPy
Statistical Methods: Cohort Analysis, Bayesian Modeling, GLMs, Hypothesis Testing, Time Series, Spatial Analysis, Causal Methods (IPW, matching, experiment design), A/B Testing, Experimental Design
ML & Data Science: Scikit-learn, TensorFlow, PyTorch, Deep Learning, NLP, LLMs, Classification, Regression, Clustering, Time-Series Analysis, Transfer Learning, Feature Engineering
Deployment & Tools: Flask, FastAPI, Docker, Git, Streamlit, Tableau, Power BI, Apache Airflow (familiar), CI/CD, Version Control
Platforms & Data Infrastructure: Linux HPC, Azure, GitHub, SQL, SQLite3, Cloud Environments (Azure, AWS — familiar), Database Technologies